

CATEGORY THEORY 4

M2000

The Adjoint Functor Theorems & monads

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The questions are divided into

- Exercises, to familiarize you with the basics
- Standard questions, which are very roughly of the kind that will make up the exam
- Further questions, which are ‘optional’ and of varying difficulty and importance.

Some questions require knowledge of other areas of mathematics; skip over any that you haven’t the background for. Please report mistakes and obscurities to leinster@dpmms.cam.ac.uk.

Exercises

1. *The essence of the Adjoint Functor Theorems is that a limit-preserving functor on a category with limits has a left adjoint. In the general case this has to be qualified by size considerations, but in the case of small categories things are cleaner.*

Let $g : Y \longrightarrow X$ be a map between (small) posets, and suppose that Y has all infima (meets, greatest lower bounds) and that g preserves them. Show that g has a left adjoint f , given by

$$f(x) = \bigwedge \{y \in Y \mid x \leq g(y)\}.$$

2. (i) What is a *monad* on a category? Explain how an adjunction gives rise to a monad, and show that every monad arises in this way.
(ii) What does it mean for a functor to be *monadic*? Show that a monadic functor is faithful and reflects isomorphisms.
3. (i) What is an *absolute-coequalizer pair* in a category $\mathcal{C}$? What does it mean for a functor $U : \mathcal{D} \longrightarrow \mathcal{C}$ to preserve coequalizers for U -absolute-coequalizer pairs, or to reflect them, or to create them?
(ii) Show that the following conditions on a functor $U : \mathcal{D} \longrightarrow \mathcal{C}$ are equivalent:
 - a. U creates coequalizers for U -absolute-coequalizer pairs
 - b. $\mathcal{D}$ has coequalizers of all U -absolute-coequalizer pairs, and U preserves and reflects such coequalizers
 - c. $\mathcal{D}$ has coequalizers of all U -absolute-coequalizer pairs, U preserves such coequalizers, and U reflects isomorphisms.
4. A *reflexive pair* is a parallel pair $X \begin{smallmatrix} \xrightarrow{p_1} \\ \xrightarrow{p_2} \end{smallmatrix} Y$ of arrows for which there exists an arrow $X \xleftarrow{t} Y$ with $p_1 t = 1_Y = p_2 t$. Show that if (T, η, μ) is a monad and $(TA \xrightarrow{\theta} A)$ a T -algebra, then $T^2 A \begin{smallmatrix} \xrightarrow{\mu_A} \\ \xrightarrow{T\theta} \end{smallmatrix} TA$ is a reflexive split-coequalizer pair.

Standard Questions

5. Consider the following three conditions on a functor U from a locally small category $\mathcal{C}$ to **Set**:
 - A.** U has a left adjoint
 - R.** U is representable
 - L.** U preserves limits.
 - (i) Show that $\mathbf{A} \Rightarrow \mathbf{R} \Rightarrow \mathbf{L}$.
 - (ii) Show that if $\mathcal{C}$ has small coproducts then $\mathbf{R} \Rightarrow \mathbf{A}$.
 - (iii) Show that if $\mathcal{C}$ is complete, well-powered and has a coseparating set then the three conditions are equivalent.

6. (i) State the General Adjoint Functor Theorem.
 - (ii) Show that if a complete, locally small category has a weakly initial set of objects then it has an initial object.
 - (iii) Explain in outline how the result in (ii) leads to a proof of the General Adjoint Functor Theorem.

7. What does it mean for a category to be *well-powered*? Show that if $\mathcal{C}$ is a well-powered category with all pullbacks then there is a functor $\text{Sub} : \mathcal{C}^{\text{op}} \longrightarrow \mathbf{Set}$ assigning to each object its set of subobjects.

*A representation of this functor is called a subobject classifier, and in fact it is possible to rephrase the definition of subobject classifier so that it makes sense in categories which aren't necessarily well-powered. A topos is a cartesian closed category with finite limits and a subobject classifier; typical examples are **Set**, $[\mathbb{C}^{\text{op}}, \mathbf{Set}]$, and the category of sheaves of sets on a fixed space.*

8. State and prove the Special Adjoint Functor Theorem. (You may assume the General Adjoint Functor Theorem.)

9. Recall from sheet 3 the theories of widgets and chads.
 - (i) Use the General Adjoint Functor Theorem to show that $U : \mathbf{Widget} \longrightarrow \mathbf{Set}$ has a left adjoint.
 - (ii) Do the same for the forgetful functor $\mathbf{Widget} \longrightarrow \mathbf{Chad}$.
 - (iii) Use the Monadicity Theorem to show that $\mathbf{Widget}$ is monadic over $\mathbf{Set}$.

10. Let (T, η, μ) be a monad on a category $\mathcal{C}$, and $U^T : \mathcal{C}^T \longrightarrow \mathcal{C}$ the forgetful functor on the category of T -algebras.
 - (i) Show that U^T creates limits.
 - (ii) Show that U^T creates coequalizers for U^T -absolute-coequalizer pairs.
 - (iii) Show that every T -algebra is a coequalizer of free T -algebras.

11. A monad (T, η, μ) is called *idempotent* if the multiplication $\mu : T^2 \longrightarrow T$ is an isomorphism. Show that every algebra for an idempotent monad is free.
12. Let $\mathbb{C}$ be a small category. The forgetful functor $U : [\mathbb{C}^{\text{op}}, \mathbf{Set}] \longrightarrow \mathbf{Set}^{\text{ob } \mathbb{C}}$ has a left adjoint F , which is given on $X \in \mathbf{Set}^{\text{ob } \mathbb{C}}$ and $C \in \mathbb{C}$ by

$$(FX)(C) = \coprod_{D \in \mathbb{C}} XD \times \mathbb{C}(C, D)$$

(as you may have discovered on sheet 2, q. 7).

- (i) Show that the adjunction is monadic.
 - (ii) Deduce that U creates limits and reflects isomorphisms. The latter statement is actually a rather familiar fact from early in the course: what?
 - (iii) Show that any presheaf of the form FX is a coproduct of representables, and deduce from a standard result on coequalizers of free algebras that every presheaf is a colimit of representables.
13. *There are many versions of the Monadicity Theorem. The following Huge Monadicity Theorem encompasses many of them, and should give you a feel for where the real substance of the Theorem lies. Part 5 would not ordinarily be included in a Monadicity Theorem, because it refers to F ; it is only included here as an intermediate step.*

Let $U : \mathcal{D} \longrightarrow \mathcal{C}$ be a functor with a left adjoint F . Prove the equivalence of the following conditions on U :

1. U is monadic
- 2a. U creates coequalizers for U -absolute-coequalizer pairs
- 2b. $\mathcal{D}$ has coequalizers of all U -absolute-coequalizer pairs, U preserves such coequalizers, and U reflects isomorphisms
- 2c. $\mathcal{D}$ has coequalizers of all U -absolute-coequalizer pairs, and U preserves and reflects such coequalizers
- 3a,b,c. As for 2, but with ‘split’ in place of ‘absolute’
- 4a,b,c. As for 3, but with ‘reflexive U -split-coequalizer’ in place of ‘ U -split-coequalizer’
5. Both of the following hold:

- (i) for each $B \in \mathcal{D}$, $FUFUB \xrightleftharpoons[\varepsilon_F U B]{FU \varepsilon_B} FUB \xrightarrow{\varepsilon_B} B$ is a coequalizer
- (ii) for each T -algebra $(TA \xrightarrow{\theta} A)$ (where T is the induced monad), there exists a map $g : FA \longrightarrow B$ in $\mathcal{D}$ such that $UFUFUA \xrightleftharpoons[UF\theta]{U\varepsilon_F A} UFA \xrightarrow{Ug} UB$ is a coequalizer.

Hints: Use q. 3(ii) to show that 2a,b,c are equivalent, and similarly 3 and 4. Then do $1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 4 \Rightarrow 5 \Rightarrow 1$.

14. This is an alternative way of presenting the Monadicity Theorem, although the methods used to prove it are just the ones we're used to.

Let $U : \mathcal{D} \longrightarrow \mathcal{C}$ be a functor with a left adjoint, T the induced monad on $\mathcal{C}$, and $K : \mathcal{D} \longrightarrow \mathcal{C}^T$ the comparison functor. Show that:

- (i) if $\mathcal{D}$ has coequalizers of U -absolute-coequalizer pairs then K has a left adjoint J
- (ii) if also U preserves such coequalizers then the unit of the adjunction $J \dashv K$ is an isomorphism
- (iii) if, in addition to this, U reflects isomorphisms, then the counit is also an isomorphism, and so U is monadic.

Further Questions

15. The objects of the category Δ are the natural numbers $0, 1, 2, \dots$, and a map $m \longrightarrow n$ is an order-preserving map from $\{0, \dots, m-1\}$ to $\{0, \dots, n-1\}$. There is a functor $\Delta \longrightarrow \mathbf{Top}$ which sends n to the standard n -dimensional simplex Δ^n , and is defined on maps via the standard face and degeneracy maps (although you don't need to know about them to do this question.) This induces a functor $S : \mathbf{Top} \longrightarrow [\Delta^{\text{op}}, \mathbf{Set}]$. Use the Special Adjoint Functor Theorem to show that S has a left adjoint.

An object of $[\Delta^{\text{op}}, \mathbf{Set}]$ is called an augmented simplicial set, and $S(X)$ is the standard 'singular complex' of a space X (from which the homotopy, homology and cohomology of X can be derived). The left adjoint to S is called geometric realization.

16. Let (T, η, μ) be a monad on a category $\mathcal{C}$, and suppose that T has a right adjoint G . Show that G has the structure of a comonad in a canonical way, and that the category of G -coalgebras is isomorphic to the category of T -algebras. (Hint: this is most efficiently done using the theory of mates—see sheet 2, q. 14, especially part (iv).)
17. True or false?
- (i) the category of fields is monadic over $\mathbf{Set}$
 - (ii) the category of posets is monadic over $\mathbf{Set}$
 - (iii) the category of totally ordered sets is monadic over $\mathbf{Set}$
 - (iv) the category of lattices is monadic over $\mathbf{Set}$ (a *lattice* being a poset with sups and infs of all finite subsets)
 - (v) the category of topological groups is monadic over $\mathbf{Top}$.